Test it honestly

AI · Level 3 · Lesson 3



Today we will

- Keep test data apart from training data, and count right and wrong answers.
- Explain overfitting: perfect on training examples, poor on new ones.
- Tell a false positive from a false negative, and say which matters more for your model.

Remember

- Put about one example in five aside: only examples the model never saw give an honest score.
- A big gap between training and test scores means overfitting. More varied data, or a simpler model, helps.
- A false positive is a false alarm; a false negative is a miss. Decide which costs more.

Memoriser and learner

1. Set up (3 min): write ten ★ sums with a secret rule.
2. Train (5 min): one memorises, one is told the rule.
3. Test (5 min): the same five sums, then five new ones.
4. Score (4 min): a training and a test score for each.
5. Explain (6 min): who overfitted, and why?

Works with three people (two players and a quizmaster) or at home with family. Keep the hidden sums out of sight until the test.

Let’s discuss

1. Why is a score on training examples not an honest test?
2. Your model is perfect on training photos and poor on new ones. What would you try?
3. For a model that spots smoke in photos to warn of fire, which mistake is worse?

Before you go

- My own example of a false alarm and a miss, and which one costs more.